

1. Count Frequency of an Element

```
ASSIGNMENT4 > G+ Q1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n, element, count = 0;
6
7      cout << "Enter size of array: ";
8      cin >> n;
9
10     int arr[n];
11
12     cout << "Enter elements: ";
13     for (int i = 0; i < n; i++)
14         cin >> arr[i];
15
16     cout << "Enter element to find frequency: ";
17     cin >> element;
18
19     for (int i = 0; i < n; i++) {
20         if (arr[i] == element)
21             count++;
22     }
23
24     cout << "Frequency of " << element << " = " << count;
25
26     return 0;
27 }
```

```
mac@SimmiTiwari ASSIGNMENT4 % /Users/mac/c++_college/ASSIGNMENT4/Q1
Enter size of array: 8
Enter elements: 1 3 6 9 7 4 6 22
Enter element to find frequency: 6
Frequency of 6 = 2
```

2. Find Largest and Smallest Element

```
ASSIGNMENT4 > G+ Q2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6
7      cout << "Enter size of array: ";
8      cin >> n;
9
10     int arr[n];
11
12     cout << "Enter elements: ";
13     for (int i = 0; i < n; i++)
14         cin >> arr[i];
15
16     int largest = arr[0];
17     int smallest = arr[0];
18
19     for (int i = 1; i < n; i++) {
20         if (arr[i] > largest)
21             largest = arr[i];
22         if (arr[i] < smallest)
23             smallest = arr[i];
24     }
25
26     cout << "Largest element = " << largest << endl;
27     cout << "Smallest element = " << smallest;
28
29     return 0;
30
31 }
```

```

● mac@SimmiTiwarei ASSIGNMENT4 % /Users/mac/c++_college/ASSIGNMENT4/Q2
Enter size of array: 5
Enter elements: 12 100 23 55 60
Largest element = 100
Smallest element = 12%

```

3. Remove Duplicate Elements

```

ASSIGNMENT4 > Q3.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6
7      cout << "Enter size of array: ";
8      cin >> n;
9
10     int arr[n];
11
12     cout << "Enter elements: ";
13     for (int i = 0; i < n; i++)
14         cin >> arr[i];
15
16     int newSize = 0;
17
18     for (int i = 0; i < n; i++) {
19         bool duplicate = false;
20
21         for (int j = 0; j < newSize; j++) {
22             if (arr[i] == arr[j]) {
23                 duplicate = true;
24                 break;
25             }
26         }
27
28         if (!duplicate) {
29             arr[newSize] = arr[i];
30             newSize++;
31         }
32     }
33
34     cout << "Array after removing duplicates: ";
35     for (int i = 0; i < newSize; i++)
36         cout << arr[i] << " ";
37
38     return 0;
39 }

```

```

● mac@SimmiTiwarei ASSIGNMENT4 % /Users/mac/c++_college/ASSIGNMENT4/Q3
Enter size of array: 5
Enter elements: 22 6 8 97 6
Array after removing duplicates: 22 6 8 97 %

```

4. Move All Zeros to the End

```

ASSIGNMENT4 > Q4.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6
7      cout << "Enter size of array: ";
8      cin >> n;
9
10     int arr[n];
11
12     cout << "Enter elements: ";
13     for (int i = 0; i < n; i++)
14         cin >> arr[i];
15
16     int position = 0;
17
18     for (int i = 0; i < n; i++) {
19         if (arr[i] != 0) {
20             arr[position] = arr[i];
21             position++;
22         }
23     }
24
25     while (position < n) {
26         arr[position] = 0;
27         position++;
28     }
29
30     cout << "Array after moving zeros to the end: ";
31     for (int i = 0; i < n; i++)
32         cout << arr[i] << " ";
33
34     return 0;
35 }

```

```

● mac@SimmiTiwari ASSIGNMENT4 % /Users/mac/c++_college/ASSIGNMENT4/Q4
Enter size of array: 5
Enter elements: 30 60 0 78 99
Array after moving zeros to the end: 30 60 78 99 0 %

```

5. Count Vowels and Consonants

```

ASSIGNMENT4 > g++ Q5.cpp > main()
1  #include <iostream>
2  #include <cctype>
3  using namespace std;
4
5  int main() {
6      string str;
7      int vowels = 0, consonants = 0;
8
9      cout << "Enter a string: ";
10     getline(cin, str);
11
12     for (char ch : str) {
13         ch = tolower(ch);
14
15         if (isalpha(ch)) {
16             if (ch == 'a' || ch == 'e' || ch == 'i' ||
17                 ch == 'o' || ch == 'u') {
18                 vowels++;
19             } else {
20                 consonants++;
21             }
22         }
23     }
24
25     cout << "Number of vowels = " << vowels << endl;
26     cout << "Number of consonants = " << consonants;
27
28     return 0;
29 }

```

```

mac@SimmiTiwari ASSIGNMENT4 % /Users/mac/c++_college/ASSIGNMENT4/Q5
Enter a string: Simmi
Number of vowels = 2
Number of consonants = 3

```